



Koneru Lakshmaiah Education Foundation

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OPERATING SYSTEMS AND SYSTEMS PROGRAMMING (25CS2104E)

2026–27, Term-I

Project Problem Statement Submission Form

Section / Team Details

Section No.: 6

Team No.: 12

Project Title: Process Termination Messages

Team Members

Roll Number	Student Name	Signature
2520030395	Monish Putti	
2520030560	Nikhil Reddy	
2520030599	Hemasai	

1. Abstract

The project focuses on implementing a Linux-based process termination message system that demonstrates how processes can be created, monitored, and terminated using Operating Systems concepts. The system provides appropriate messages when a process terminates, helping users understand the process lifecycle and termination status. The project uses Linux process management mechanisms and system programming concepts to handle process creation, execution, and termination. It demonstrates the interaction between parent and child processes and the use of termination-related system calls and signals. The proposed solution is implemented in C/C++ on a Linux environment such as Ubuntu. The system aims to provide a simple and clear demonstration of process termination and communication between processes. The expected outcome is a functional Linux program that displays meaningful termination messages and provides practical understanding of process management, process states, signals, and system calls. This project helps connect theoretical Operating Systems concepts with their practical implementation in a Linux environment.

2. Problem Statement

In a Linux operating system, processes may terminate normally or due to signals and other termination conditions. Understanding how a parent process detects and responds to the termination of a child process is an important Operating Systems concept. The project aims to develop a Linux-based system that demonstrates process termination and displays appropriate termination messages. The system will use process management and system programming mechanisms to create processes,

monitor their execution, detect their termination status, and communicate the result through meaningful messages.

3. Objectives

1. To create and manage processes in a Linux environment.
2. To demonstrate the termination of child processes and detection of their termination status.
3. To use Linux system calls, signals, and process management mechanisms for handling process termination.
4. To display meaningful termination messages and demonstrate practical Operating Systems concepts.

3. Proposed Methodology

The project will be designed and implemented in C/C++ on Linux/Ubuntu. The program will create and manage processes using Linux process management mechanisms. A parent process will create a child process and monitor its execution. The child process will perform a specified operation and terminate either normally or through a termination signal. The parent process will wait for the child process and obtain its termination status using appropriate Linux system calls. Based on the termination condition, the program will display a suitable process termination message. The implementation will demonstrate process creation, parent-child process interaction, process termination, signals, waiting, and termination-status handling.

5. Operating Systems Concepts / Linux APIs Used

OS Concept / Linux API / System Call	Purpose in the Project
fork()	Creates a child process from the parent process.
wait() / waitpid()	Allows the parent process to wait for and obtain the termination status of the child process.
exit()	Terminates a process normally with an exit status.
kill()	Sends a signal to a process for demonstrating signal-based termination.
Signals	Used to handle and demonstrate process termination caused by signals.

6. Individual Contribution

Roll Number	Student Name	Individual Responsibility
2520030395	Monish Putti	1) Process management implementation and integration of the project modules. 2) Documentation, testing, result verification, and project presentation support.
2520030560	Nikhil Reddy	1) Implementation and testing of process creation and termination mechanisms.
2520030599	Hemasai	1) Implementation of termination messages, signals, and process-status handling.

7. Tools / Platforms / Software Used

Tool / Platform / Software	Purpose
Linux / Ubuntu	Provides the operating system environment for implementing and executing the project.
C / C++	Used to implement process creation, termination, signals, and process management.
GCC Compiler	Compiles the C/C++ source code and generates the executable program.
Bash / Linux Terminal	Used to compile, execute, and test the program.
GitHub	Used for source-code version control and project repository management.

8. Expected Outcome

The project is expected to produce a functional Linux-based program that demonstrates process termination and displays appropriate termination messages. The system will create and manage parent and child processes, monitor the child process, detect its termination status, and display whether the process terminated normally or due to a signal. The implementation will provide practical understanding of process management, process states, signals, system calls, and parent-child process interaction in Linux.

9. GitHub Repository

GitHub Repository Link: <https://github.com/dumpalahemasai07/OS--PROJECT>

Repository Created: _____

☒ Yes ☐ No

Faculty Name and Signature

Faculty Name: Dr. K. Anusha

Signature:

Remarks:

Faculty Use

Project Approved: _____

☐ Yes ☐ No

Date: _____

Faculty Signature: _____